

Cube Test Report — Pier P5 Pier Shaft Pour, 14-Feb-2022

Nandira River Bridge Project — Package NRB-4

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Date: 15-Mar-2022
Contract No.: NHIA/NRB4/CW/2020-01
From: Suvarna Materials Testing Laboratory
To: Meridian Engineering Consultants
CC: Sagara Constructions Pvt. Ltd.

Subject: Compressive Strength Test Results — Pier P5 Pier Shaft, Pour Date 14-Feb-2022, Grade M40

Specified characteristic strength: 40 N/mm² at 28 days (Grade M40, per Employer's Requirements Section 5 and the Specification).

TEST SET A (first ~30m³, cast 09:15–10:50):

7-day results (individual): 26.8, 28.1, 27.6 N/mm² — Average: 27.5 N/mm²

28-day results (individual): 42.1, 43.5, 41.8 N/mm² — Average: 42.5 N/mm²

ASSESSMENT: Satisfactory. Exceeds specified characteristic strength.

TEST SET B (middle ~30m³, cast 11:40–13:10):

7-day results (individual): 24.2, 25.6, 24.6 N/mm² — Average: 24.8 N/mm²

28-day results (individual): 35.2, 33.8, 36.1 N/mm² — Average: 35.0 N/mm²

ASSESSMENT: Does not meet specified characteristic strength. Both the average and each individual result fall below the acceptance criteria in the Specification.

TEST SET C (final ~2m³, cast 13:10–13:25):

7-day results (individual): 27.4, 28.9, 28.0 N/mm² — Average: 28.1 N/mm²

28-day results (individual): 44.0, 42.7, 43.1 N/mm² — Average: 43.3 N/mm²

ASSESSMENT: Satisfactory. Exceeds specified characteristic strength.

Cubes cured in the site laboratory curing tank in accordance with the Specification, save that a fault in the curing tank heater was logged by the Engineer between approximately 18-Feb-2022 and 04-Mar-2022 (repaired 04-Mar-2022); this period overlaps the early curing age of the 28-day cubes for this pour.

Issued by: Suvarna Materials Testing Laboratory